

Gas-specific precision and vertical concentration gradients across spatio-temporal resolutions in a naturally ventilated dairy barn

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Abstract

Concentration measurements in naturally ventilated dairy barns are often interpreted as though spatial and temporal averaging affected all gases similarly. This assumption was tested by measuring carbon dioxide (CO₂), methane (CH₄), and ammonia (NH₃) at 51 internal sampling points arranged as 17 vertical triplets during Campaign 1, followed by a 32-point top-and-bottom network during Campaign 2. Two cavity ring-down spectrometers and, in Campaign 1, two Fourier-transform infrared spectrometers sampled sequentially at four-minute resolution. Fourier-transform infrared concentrations were harmonised to the cavity ring-down scale after a 24-h co-location test. Precision was quantified as the spatial coefficient of variation at hourly, daily, weekly, and whole-period resolutions. Linear mixed-effects models tested height, period, and their interaction; gas ratios were analysed to distinguish gas-specific gradients from common dilution. Precision depended strongly on gas and temporal aggregation (response-by-scale interaction, $p < 0.001$). Median hourly spatial coefficients of variation were approximately 7.5–8.9% for CO₂, 22.0–27.2% for CH₄, and 24.3–33.2% for NH₃, and generally decreased at daily and weekly scales. Height effects and height-by-period interactions were significant for all concentrations and ratios ($p < 0.001$). Top concentrations exceeded bottom concentrations most consistently in autumn; in summer, the vertical increase was weaker and not universal across horizontal positions. Thus, the hypothesis of gas-specific spatio-temporal precision was supported, whereas the hypothesised general increase with sampling height was seasonally conditional. The middle height, 3.6 m above the floor and only 1 m above the bottom points, added comparatively little vertical

separation during summer, supporting its omission as a practical consequence of the heterogeneity analysis rather than as evidence of homogeneous mixing.

Keywords: ammonia; dairy housing; methane; mixing ratio; sampling height; spatial coefficient of variation

Nomenclature

Symbol	Definition
CRDS	cavity ring-down spectroscopy
CV	coefficient of variation, %
FTIR	Fourier-transform infrared spectroscopy
NVDB	naturally ventilated dairy barn
RE	relative error against the contemporaneous dense-network mean, %
SD	standard deviation
SP	sampling point

1 Introduction

Emission estimates for naturally ventilated livestock buildings depend jointly on airflow and the concentration field sampled across that airflow. Wind, buoyancy, animals, manure surfaces and internal obstructions produce spatially variable transport, so analytical accuracy alone does not ensure spatial representativeness (Calvet et al., 2013; Ogink et al., 2013). Direct wind-tunnel measurements demonstrated that combining concentration and velocity at one constant height could produce emission errors up to 97%, whereas vertically resolved sampling reduced the error to below 5% in the tested cases (Janke et al., 2020b). A concentration sensor should therefore be evaluated at the spatial and temporal resolution at which it will be used.

Vertical gradients are gas- and site-specific. Sahu et al. (2023) measured significant height effects for CO₂, CH₄, and NH₃ at the outlet of a naturally ventilated dairy building and commonly observed greater concentrations near the top. In another open dairy barn, NH₃ was greatest near the floor whereas CH₄ was greatest at the upper level, demonstrating that a universal vertical ordering cannot be assumed (D’Urso et al., 2024). Horizontal sampling density also affects inferred means and variability (D’Urso et al., 2022; De Voeleer et al., 2017). Dombia et al. (2021) further showed that the frequently used approximately 3 m sampling height represented well-mixed conditions only for a subset of simulated convection cases. These results motivate measurements both near the customary height and at uncommon vertical extremes.

Temporal aggregation is equally important. Rapid changes in wind and thermal forcing may create large short-term spatial differences that are attenuated in daily or campaign means. Wind direction can also change which openings are inlets or outlets and which indoor locations are representative (Saha et al., 2013; König et al., 2018; Janke et al., 2020a). Regional meteorological observations describe the synoptic forcing but do not replace velocity measurements at the barn openings. Consequently, seasonal comparisons should quantify temporal evolution while avoiding causal claims that cannot be resolved without co-located airflow measurements.

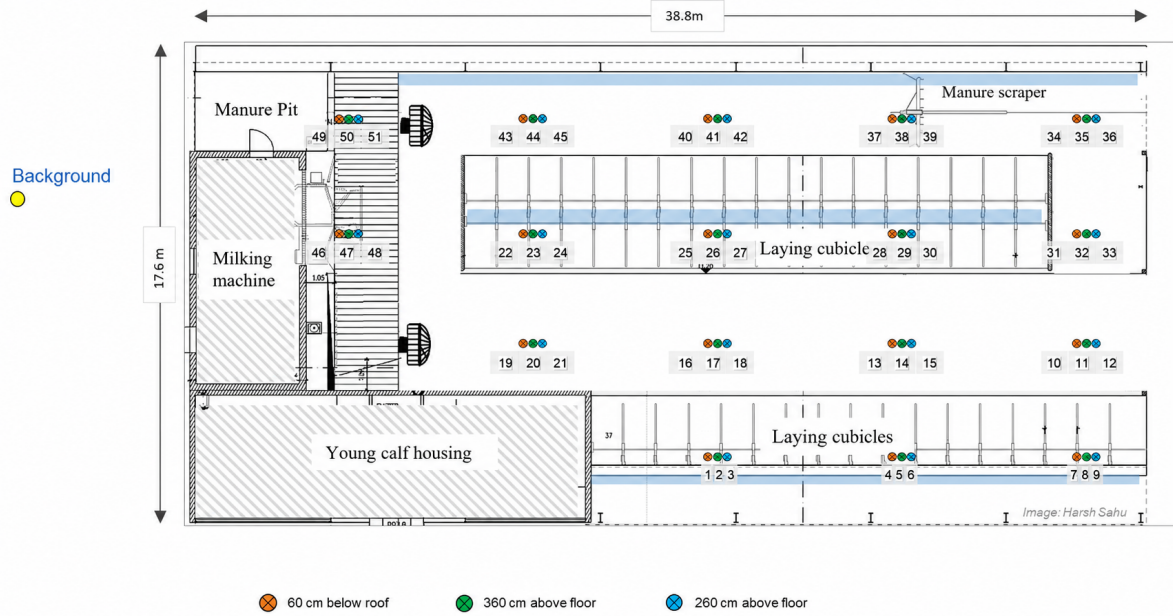
Gas ratios provide a complementary diagnostic. If all gases respond identically to dilution, their ratios should be spatially more stable than their concentrations. Persistent gradients in NH_3/CO_2 , CH_4/CO_2 , or NH_3/CH_4 instead indicate differences in sources, sinks or transport. Chen et al. (2026) used entropy to describe the sensitivity of CFD airflow fields to prescribed wind cases. Here, entropy instead describes temporal state occupancy in sequential field concentrations and is interpreted as a complementary, gas-specific measure of location informativeness.

Three hypotheses were tested: (H1) the precision of each gas differs at different spatial and temporal resolutions; (H2) gas concentrations generally increase with increasing vertical sampling height, as suggested by previous measurements (Sahu et al., 2023); and (H3) the centrally located SP25 is equivalent to the daily whole-network mean within a prespecified $\pm 5\%$ relative-error margin. The objectives were to quantify concentration and ratio heterogeneity, test the persistence of vertical structure, evaluate SP25 as a pragmatic reference, and assess omission of the middle layer as a practical consequence of the observed structure.

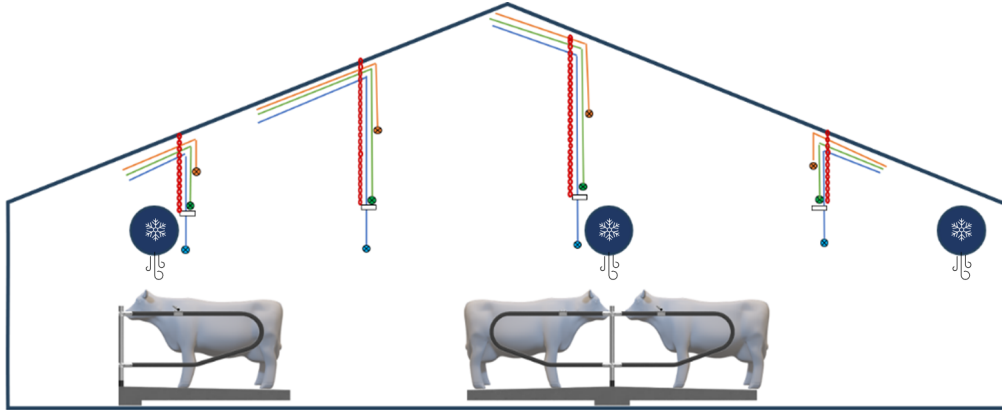
2 Materials and methods

2.1 Barn and sampling layout

Measurements were conducted in an occupied NVDB at Gross Kreutz, Brandenburg, Germany. The barn was 38.88 m long and 17.65 m wide, with an internal airspace of approximately 4529 m^3 and capacity for 58 dairy cows. Seventeen horizontal positions covered lying, feeding, walking and manure-adjacent zones (Fig. 1). At every position, a vertical triplet comprised a top SP 0.60 m below the local roof, a middle SP 3.60 m above the floor, and a bottom SP 2.60 m above the floor. Thus, the middle and bottom layers were separated by only 1.00 m; the top layer followed the sloping roof and represented an intentionally uncommon near-roof position.



(a) Plan view and 51 internal SPs.



(b) Vertical arrangement of the sampling tubes.

Figure 1: Experimental barn and sampling design. Location numbers increase in vertical triplets: top, middle and bottom. The plan also shows functional zones and mechanical mixing fans.

2.2 Campaigns and gas measurements

Campaign 1 combined dense observations from June–August 2024 (summer; 90 measurement dates) and 1–24 October 2024 (autumn; 24 dates). All 51 internal SPs were measured with two CRDS and two FTIR analysers. Campaign 2 used the reduced top-and-bottom design, measured by two CRDS analysers from 16 November to 31 December 2024 (15 late-autumn and 31 winter dates). Thirty-two positions were operational because two previously identified positions were omitted for pragmatic reasons. Sampling proceeded sequentially at four-minute resolution. The first 60 s after switching was discarded and the remaining 180 s were averaged; the time stamp represented the end of the averaging interval.

A preliminary 24-h co-located field comparison was performed before Campaign 1. Two polytetrafluoroethylene sampling lines terminated at the same central barn position; one line was divided between the CRDS instruments and the other between the FTIR instruments. FTIR values were 6% higher for CO₂ and CH₄, and 9% higher for NH₃. Campaign 1 FTIR values were therefore harmonised as $C_{\text{CO}_2,\text{corr}} = C_{\text{CO}_2,\text{raw}}/1.06$, $C_{\text{CH}_4,\text{corr}} = C_{\text{CH}_4,\text{raw}}/1.06$, and $C_{\text{NH}_3,\text{corr}} = C_{\text{NH}_3,\text{raw}}/1.09$. CRDS values were unchanged. Only missing, zero or negative gas values were excluded; no lower positive-concentration threshold was imposed.

2.3 Meteorological context

Hourly temperature, relative humidity, precipitation, wind speed and wind direction were obtained from the German Meteorological Service for the nearest suitable station and aligned in UTC (Deutscher Wetterdienst Climate Data Center, 2024). Wind roses and temperature–humidity time series were used to describe seasonal forcing. Westerly flow was defined as 225–315°. Hourly internal concentration means under westerly and other directions were compared using Wilcoxon tests, and exploratory mixed models included wind speed, direction class, temperature, relative humidity and precipitation. These analyses describe association, not barn-scale causation, because velocity and direction were not measured simultaneously at every opening.

2.4 Responses and statistical analysis

Ratios were calculated on paired observations as

$$R_{a/b} = C_a/C_b. \quad (1)$$

Spatial precision within each hour, day, ISO week and campaign period was expressed as

$$CV_{g,t} = 100 \frac{SD_i(C_{g,i,t})}{\overline{C}_{g,t}}, \quad (2)$$

where i is SP and t is aggregation interval. Because ratio distributions can be sensitive to small denominators, medians and interquartile ranges of block-level CVs were emphasised alongside campaign means and SDs. A mixed model tested log-transformed CV with response, temporal scale and their interaction as fixed effects and period as a random intercept.

For dense Campaign 1 observations, log response was modelled with height, period and their interaction as fixed effects and horizontal position and date as crossed random intercepts. A second model used only locations shared by both campaigns and tested top versus bottom across four periods, again with horizontal position and date as random effects. Tukey-adjusted estimated-marginal-mean contrasts were calculated. The consistency of H2 was further expressed as the number of horizontal positions at which $\overline{C}_{\text{top}} > \overline{C}_{\text{middle}} > \overline{C}_{\text{bottom}}$.

For complete two-hour dense-network blocks, each single-height mean was compared with the 51-location mean:

$$RE_{h,b} = 100 \frac{\overline{C}_{h,b} - \overline{C}_{\text{dense},b}}{\overline{C}_{\text{dense},b}}. \quad (3)$$

Shannon entropy was calculated after discretising each response into four equiprobable bins within period. The resulting tiles assess empirical compositional information at a location; they are not estimates of airflow or local mean age of air. Analyses were conducted in R 4.5.3. Significance was assessed at $\alpha = 0.05$, with effect sizes and temporal consistency interpreted alongside p -values.

SP25 was compared with the equally weighted daily whole-network mean using signed RE, mean absolute error, RMSE and Spearman correlation. Equivalence was assessed using the two one-sided-test principle: the 90% confidence interval for mean daily RE had to lie completely within -5% to $+5\%$. This margin was specified before interpretation and was applied identically to concentrations and ratios.

3 Results

3.1 Gas-specific precision across temporal resolutions

The analysis included 159,866 internal records: 108,078 in summer Campaign 1, 31,156 in autumn Campaign 1, 8,906 in late-autumn Campaign 2 and 11,726 in winter Campaign 2. Precision differed by response, temporal scale and their interaction (all $p < 0.001$). At

hourly resolution, median spatial CV was 7.5–8.9% for CO_2 , 22.0–27.2% for CH_4 , and 24.3–33.2% for NH_3 . Daily medians decreased to 4.4–6.1%, 9.5–17.1%, and 19.9–29.3%, respectively. Weekly values were similar to or below daily values for CO_2 and CH_4 , whereas NH_3 remained comparatively heterogeneous. H1 was therefore supported: both gas identity and aggregation scale altered apparent precision.

Mean concentrations also changed over time. Across heights, CO_2 and CH_4 were generally higher in autumn and winter than in summer, while NH_3 declined from approximately 2.1 ppm in summer to below 1 ppm in Campaign 2. The ratios did not track concentration changes identically, confirming that temporal variation was not a uniform multiplication of all gases.

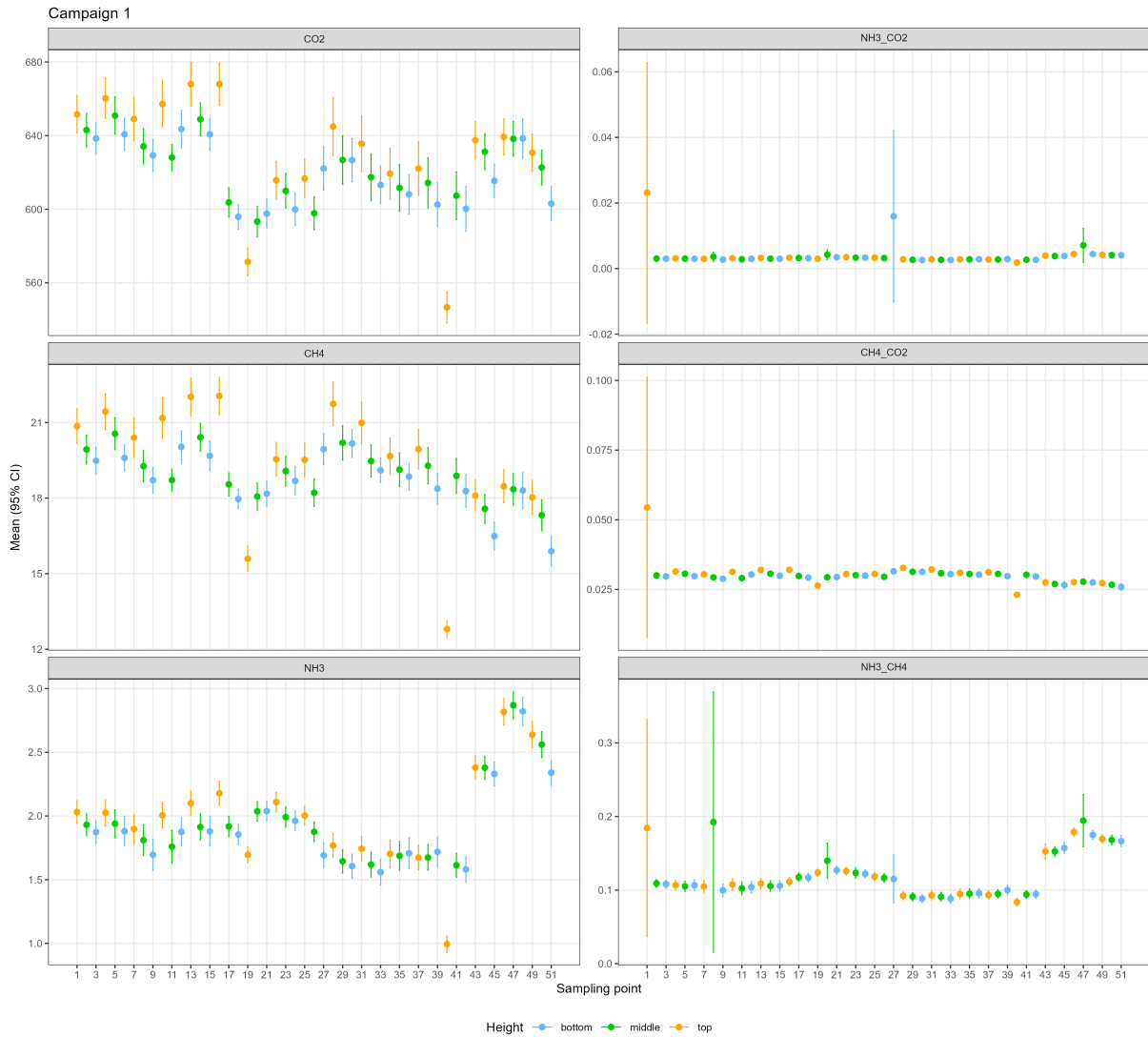


Figure 2: Campaign 1 mean concentrations and dimensionless ratios with 95% confidence intervals calculated from daily sampling-point means. All 51 internal SPs are shown. Facets follow the common response order; colours denote top (orange), middle (green), and bottom (blue).

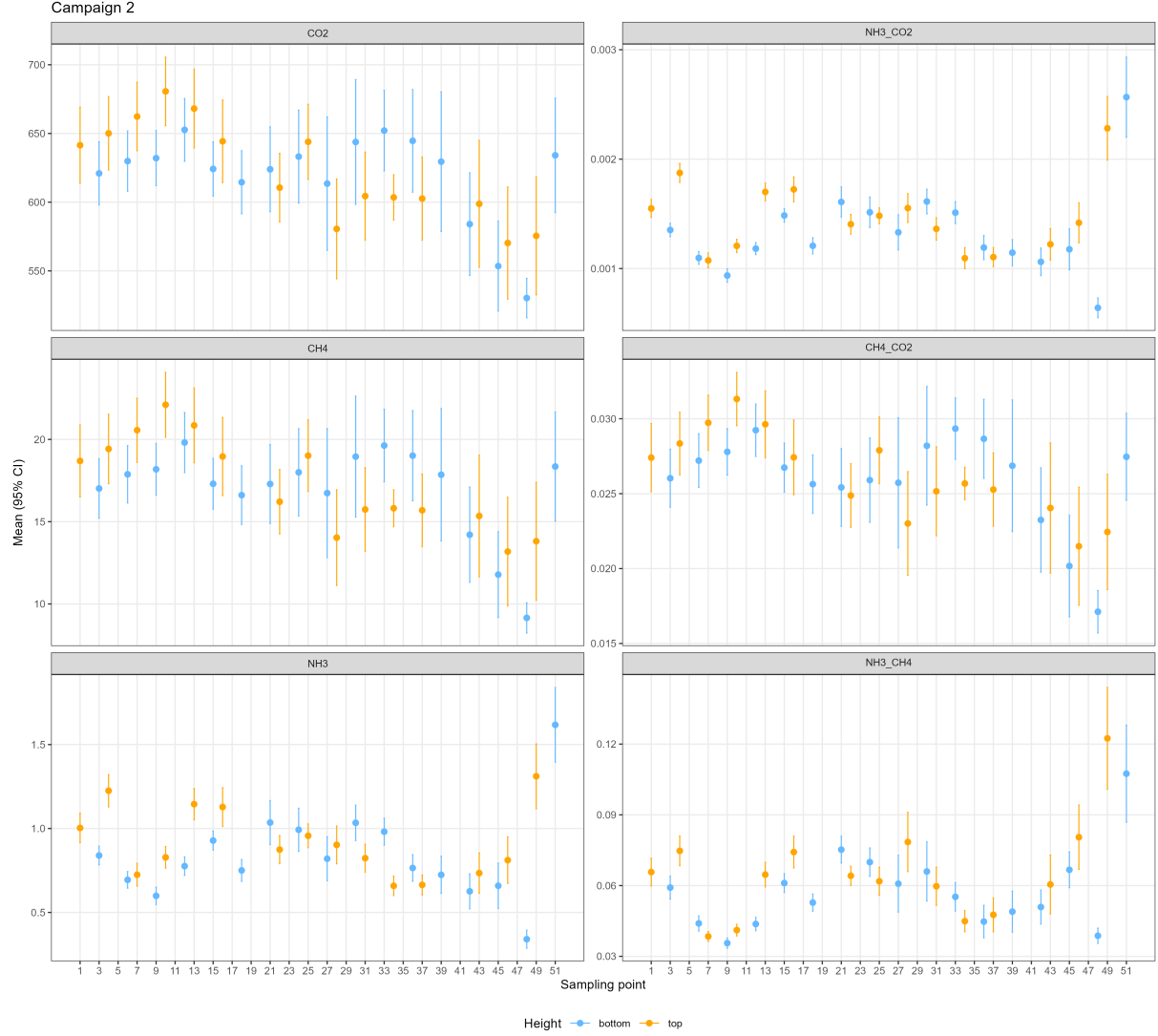
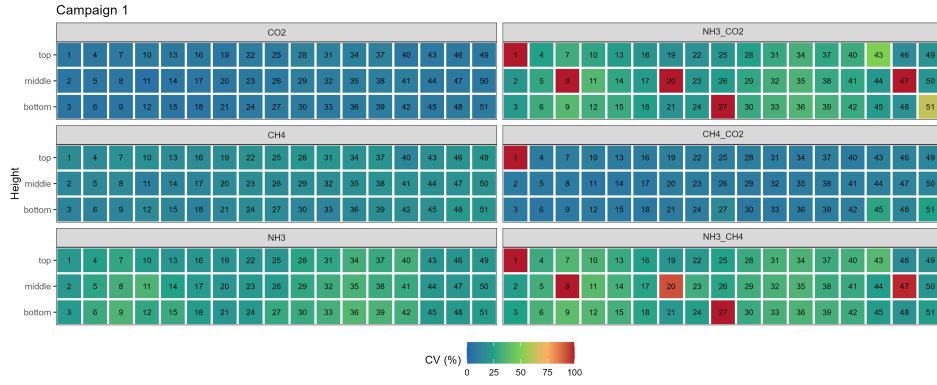
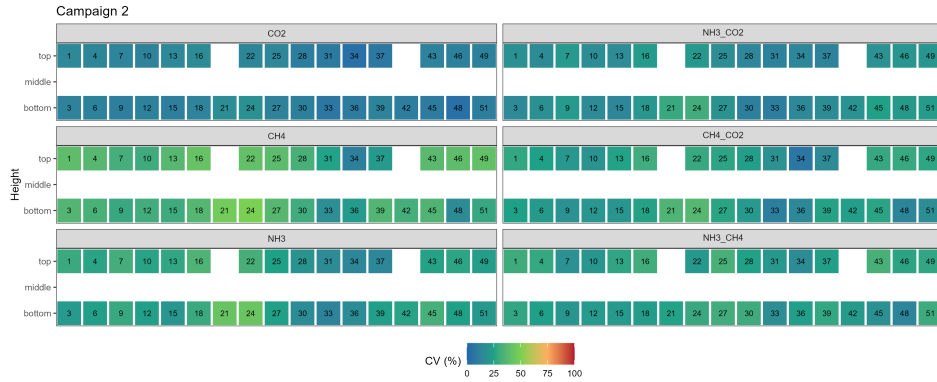


Figure 3: Campaign 2 mean concentrations and dimensionless ratios with 95% confidence intervals calculated from daily sampling-point means. The reduced network contains top (orange) and bottom (blue) SPs; no middle SPs were sampled.



(a) Campaign 1.



(b) Campaign 2.

Figure 4: Location-specific coefficients of variation calculated from daily means. All panels use the same 0–100% colour scale; location numbers remain inside the tiles.

3.2 Vertical structure and its seasonal dependence

In dense Campaign 1 data, height, period and their interaction affected every concentration and ratio (all interaction $p < 0.001$). Summer gradients were small: geometric top means exceeded bottom means by approximately 0.9% for CO₂, 1.1% for CH₄, and 1.6% for NH₃. In autumn, corresponding contrasts increased to approximately 4.7%, 10.8%, and 15.9%. The strict top–middle–bottom ordering occurred at only 9/17, 10/17, and 10/17 positions for the three gases in summer, but at 13/17, 13/17, and 14/17 in autumn. H2 was therefore conditionally, not universally, supported.

The four-period top–bottom models confirmed significant height-by-period interactions for all responses ($p < 0.001$). Top concentrations exceeded bottom concentrations significantly in every period except CO₂ in late autumn. The largest proportional separation occurred for NH₃, especially in autumn and winter. Ratio gradients were also significant, showing that common ventilation-driven dilution alone could not explain the vertical ordering.

The middle-only distribution generally lay closer to the dense mean than either extreme, as expected for an intermediate layer. Its practical incremental value was nevertheless limited during summer because it was only 1 m above the bottom layer and frequently indistinguishable from either bottom or top for particular gases. This empirical redundancy motivated its omission in Campaign 2; it does not imply that the barn was homogeneously mixed.

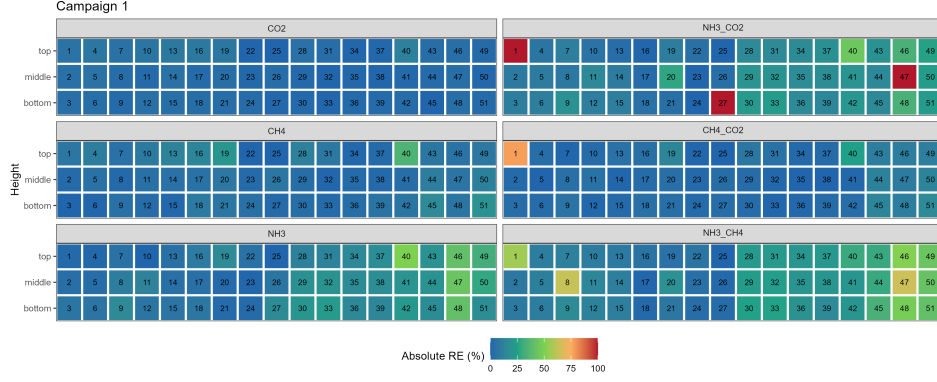
3.3 Representativeness of the central SP25

SP25 reproduced daily whole-network CO₂ particularly well: mean absolute RE was 2.16% in Campaign 1 and 1.99% in Campaign 2, RMSE was 2.70% and 2.47%, and Spearman correlation was 0.959 and 0.987, respectively. Performance was weaker for CH₄ (mean absolute RE 6.02–6.63%) and NH₃ (7.68–10.43%). Ratio performance was also response-dependent; Campaign 2 NH₃/CO₂ had 6.86% signed bias and 10.18% mean absolute RE. Consequently, H3 was supported for CO₂ but not as a universal statement across gases and ratios. Central position alone was insufficient to establish a generally representative sampling point.

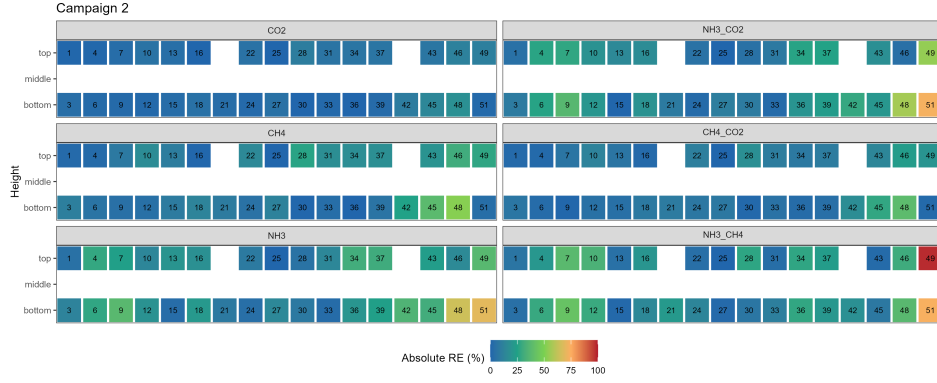
3.4 Entropy and meteorological context

Entropy varied by gas and position (Fig. 6). Locations with high entropy for one gas were not invariably high for another, consistent with gas-specific source and transport behaviour. The result did not identify a universal sensor position and should not be equated with CFD-based optimisation.

Seasonal wind roses and temperature–humidity records showed that the campaigns



(a) Campaign 1.



(b) Campaign 2.

Figure 5: Absolute relative error of each location mean against the pragmatic central reference, SP 25. Ratios were calculated without percentage scaling; only their relative errors are percentages. All panels use the same 0–100% colour scale.

Table 1: Performance and equivalence of SP25 against the equally weighted daily whole-network mean. Equivalence required the 90% confidence interval (CI) for signed relative error (RE) to lie entirely within -5 to $+5\%$.

Campaign	Response	Signed RE (%)	90% CI (%)	MAE (%)	RMSE (%)	ρ	Equivalent $\pm 5\%$
Campaign 1	CO ₂	-0.56	-0.97 to -0.14	2.16	2.70	0.959	Yes
Campaign 1	NH ₃ /CO ₂	0.98	-1.28 to 3.24	8.84	14.52	0.876	Yes
Campaign 1	CH ₄	4.09	3.08 to 5.09	6.02	7.63	0.945	No
Campaign 1	CH ₄ /CO ₂	4.13	2.97 to 5.29	5.83	8.50	0.891	No
Campaign 1	NH ₃	1.02	-0.42 to 2.46	7.68	9.29	0.880	Yes
Campaign 1	NH ₃ /CH ₄	-3.39	-5.45 to -1.32	9.55	13.64	0.849	No
Campaign 2	CO ₂	0.74	0.14 to 1.33	1.99	2.47	0.987	Yes
Campaign 2	NH ₃ /CO ₂	6.86	4.39 to 9.33	10.18	12.02	0.826	No
Campaign 2	CH ₄	1.55	-0.44 to 3.55	6.63	8.12	0.983	Yes
Campaign 2	CH ₄ /CO ₂	1.25	-0.18 to 2.68	4.83	5.84	0.987	Yes
Campaign 2	NH ₃	7.28	4.64 to 9.92	10.43	12.80	0.886	No
Campaign 2	NH ₃ /CH ₄	4.61	1.79 to 7.43	9.53	12.18	0.956	No

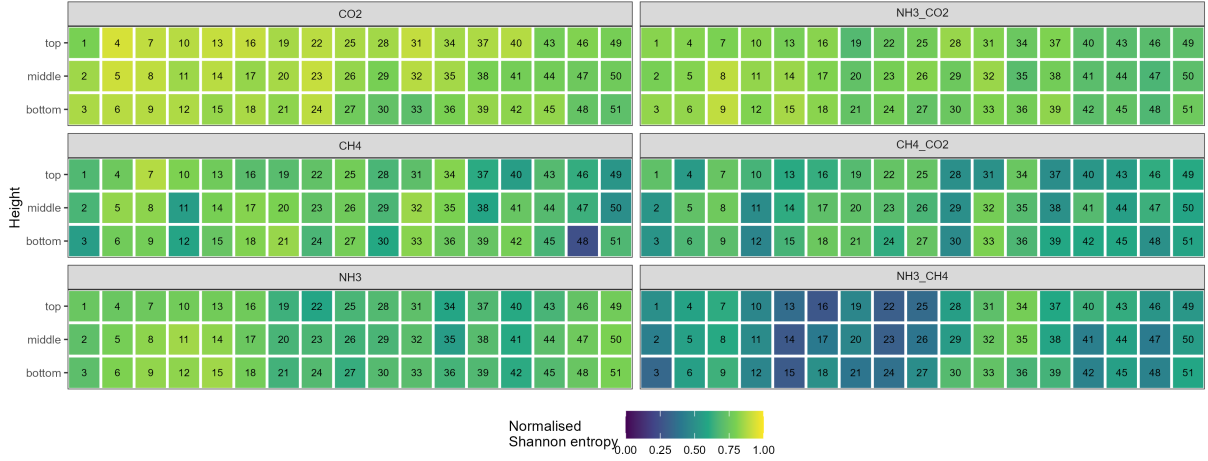


Figure 6: Four-bin Shannon entropy for each gas and ratio at horizontal positions 1–17 and the three heights during Campaign 1. Equal coordinate scaling produces square tiles. Higher entropy denotes a more even occupancy of the four concentration bins, not lower measurement uncertainty.

sampled different external regimes (Fig. 7). Westerly hours were associated with lower, rather than higher, internal hourly mean concentrations in the unadjusted summaries: for example, autumn Campaign 1 means changed from 671 to 601 ppm CO₂, 21.3 to 17.2 ppm CH₄, and 1.76 to 1.37 ppm NH₃. Winter differences were also negative. These associations are consistent with enhanced ventilation under some westerly regimes, but regional wind cannot distinguish dilution from plume transport or opening-specific flow.

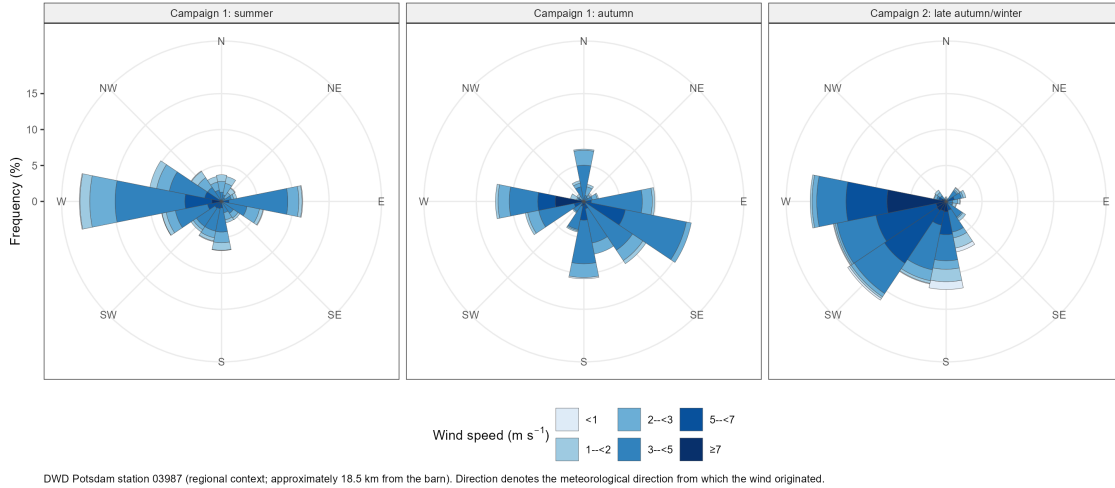


Figure 7: Seasonal wind direction and speed from the German Meteorological Service. These regional measurements describe seasonal forcing and were not treated as direct measurements of airflow through the barn.

4 Discussion

4.1 Precision is a property of gas, space and time

The first hypothesis was strongly supported. A statement such as “the concentration was spatially variable” is incomplete unless gas and aggregation interval are specified. CO₂ had the lowest hourly CV, consistent with a broadly distributed animal source and a relatively large ambient baseline. NH₃ and CH₄ retained greater spatial variability, consistent with local manure, animal and activity sources. Aggregation reduced much of the transient variability but did not eliminate gas-specific differences. This explains how campaign means may appear stable while hourly sensor placement remains consequential for emission calculations.

The result complements the direct emission analysis of Janke et al. (2020b): their large error at one constant height arose from jointly unresolved concentration and velocity profiles, whereas the present work isolates the concentration component in an occupied barn. The decrease in CV with aggregation should therefore not be interpreted as proof that a temporally averaged single concentration is flux-representative. Concentration and velocity must ultimately be sampled at matching spatial and temporal resolutions.

Table 2: Descriptive statistics for daily whole-network estimates and sampling point (SP) 25. Gas ratios are dimensionless and are not multiplied by 100. Whole-network daily estimates weight each available SP equally. Relative error (RE) is signed for SP 25 against the contemporaneous network mean.

CO ₂					
Statistic	C1 network	C1 SP25	C2 network	C2 SP25	
Mean	619.66	616.75	638.52	643.93	
SD	47.11	56.01	84.67	91.76	
CV (%)	7.60	9.08	13.26	14.25	
Median	615.25	610.67	640.47	652.74	
MAD	36.86	40.40	76.15	69.82	
Minimum	501.60	470.99	489.69	484.90	
Q1	582.23	575.34	562.83	555.29	
Q3	658.40	663.17	711.44	713.82	
Maximum	740.94	767.14	797.24	834.89	
95% CI lower	610.92	606.36	613.38	616.68	
95% CI upper	628.41	627.15	663.67	671.18	
Daily estimates	114	114	46	46	
Measurement days	114	114	46	46	
SP25 RE (%)	0.00	-0.56	0.00	0.74	

CH ₄					
Statistic	C1 network	C1 SP25	C2 network	C2 SP25	
Mean	18.70	19.52	18.53	19.01	
SD	2.87	3.58	6.65	7.33	
CV (%)	15.33	18.34	35.86	38.56	
Median	18.49	19.17	18.55	19.69	
MAD	2.08	2.58	5.80	6.07	
Minimum	9.73	9.12	7.29	6.79	
Q1	16.64	16.78	12.26	12.45	
Q3	20.83	21.98	24.07	24.35	
Maximum	27.37	30.12	30.86	34.32	
95% CI lower	18.17	18.86	16.56	16.84	
95% CI upper	19.23	20.19	20.51	21.19	
Daily estimates	114	114	46	46	
Measurement days	114	114	46	46	
SP25 RE (%)	0.00	4.09	0.00	1.55	

NH ₃					
Statistic	C1 network	C1 SP25	C2 network	C2 SP25	
Mean	1.99	2.00	0.90	0.96	
SD	0.40	0.39	0.25	0.24	
CV (%)	19.98	19.60	28.08	24.83	
Median	2.01	1.99	0.84	0.90	
MAD	0.27	0.26	0.16	0.15	
Minimum	0.89	0.92	0.53	0.60	
Q1	1.70	1.72	0.73	0.79	
Q3	2.23	2.25	1.04	1.09	
Maximum	3.23	3.08	1.63	1.58	
95% CI lower	1.92	1.93	0.83	0.89	
95% CI upper	2.07	2.08	0.98	1.03	
Daily estimates	114	114	46	46	
Measurement days	114	114	46	46	
SP25 RE (%)	0.00	1.02	0.00	7.28	

NH ₃ /CO ₂					
Statistic	C1 network	C1 SP25	C2 network	C2 SP25	
Mean	0.00386	0.00332	0.00139	0.00148	
SD	0.00580	0.00068	0.00024	0.00024	
CV (%)	150.36	20.56	17.51	16.38	
Median	0.00331	0.00327	0.00134	0.00143	
MAD	0.00050	0.00052	0.00013	0.00015	
Minimum	0.00158	0.00166	0.00108	0.00112	
Q1	0.00273	0.00284	0.00123	0.00129	
Q3	0.00375	0.00389	0.00150	0.00160	
Maximum	0.06441	0.00483	0.00212	0.00206	
95% CI lower	0.00278	0.00319	0.00132	0.00141	
95% CI upper	0.00494	0.00344	0.00147	0.00155	
Daily estimates	114	114	46	46	
Measurement days	114	114	46	46	
SP25 RE (%)	0.00	0.98	0.00	6.86	

CH ₄ /CO ₂					
Statistic	C1 network	C1 SP25	C2 network	C2 SP25	
Mean	0.02960	0.03059	0.02740	0.02790	
SD	0.00506	0.00322	0.00679	0.00749	
CV (%)	17.10	10.53	24.78	26.86	
Median	0.02956	0.03054	0.02772	0.02897	
MAD	0.00156	0.00209	0.00557	0.00610	
Minimum	0.01906	0.01916	0.01457	0.01376	
Q1	0.02777	0.02852	0.02117	0.02117	
Q3	0.03098	0.03264	0.03294	0.03375	
Maximum	0.07539	0.03916	0.03792	0.04021	
95% CI lower	0.02866	0.02999	0.02539	0.02567	
95% CI upper	0.03054	0.03119	0.02942	0.03012	
Daily estimates	114	114	46	46	
Measurement days	114	114	46	46	
SP25 RE (%)	0.00	4.13	0.00	1.25	

NH ₃ /CH ₄					
Statistic	C1 network	C1 SP25	C2 network	C2 SP25	
Mean	0.12497	0.11828	0.05816	0.06182	
SD	0.03718	0.02986	0.01401	0.02006	
CV (%)	29.75	25.24	24.10	32.45	
Median	0.12621	0.11830	0.05723	0.05714	
MAD	0.02288	0.02323	0.00968	0.01476	
Minimum	0.07046	0.05895	0.03216	0.03065	
Q1	0.09719	0.09451	0.04761	0.04473	
Q3	0.14548	0.13849	0.06684	0.07464	
Maximum	0.30039	0.18517	0.08961	0.11027	
95% CI lower	0.11807	0.11274	0.05400	0.05587	
95% CI upper	0.13187	0.12382	0.06232	0.06778	
Daily estimates	114	114	46	46	
Measurement days	114	114	46	46	
SP25 RE (%)	0.00	-3.39	0.00	4.61	

Table 3: Omnibus tests from the linear mixed-effects models. The dense model used Campaign 1 and all three heights; the shared model used top and bottom SPs available across the four seasonal periods. Models were fitted to log-transformed responses with horizontal position and date as random intercepts.

Response	Campaign 1 dense			Shared top–bottom		
	Height	Period	Height $\times$ period	Height	Period	Height $\times$ period
CO ₂	< 0.001	< 0.001	< 0.001	< 0.001	0.002	< 0.001
NH ₃ /CO ₂	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
CH ₄	< 0.001	< 0.001	< 0.001	< 0.001	0.008	< 0.001
CH ₄ /CO ₂	< 0.001	0.053	< 0.001	< 0.001	< 0.001	< 0.001
NH ₃	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
NH ₃ /CH ₄	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001

4.2 The vertical-increase hypothesis was conditional

The near-roof increase reported by Sahu et al. (2023) was reproduced most clearly in autumn, but only weakly in summer and not at every horizontal position. This partial support is scientifically more informative than a universal confirmation. It agrees with the gas-specific profiles of D’Urso et al. (2024) and with the simulation result of Doumbia et al. (2021) that an approximately 3 m line represents only some flow regimes. Roof geometry, buoyant animal plumes, fan operation, wind-driven short-circuiting and local sources can each alter the vertical ordering.

The middle SPs at 3.6 m were close to the common measurement height and only 1 m above the bottom SPs. The dense design was valuable because it bracketed that conventional region with a bottom layer at 2.6 m and an uncommon near-roof layer. In summer, the limited middle–bottom separation and small overall gradients made the middle layer comparatively redundant. Its removal in Campaign 2 was therefore a pragmatic outcome of measured heterogeneity. It should not be generalised to barns with different roof slopes, opening geometry, manure management or mechanical air mixing.

Ratio gradients reinforce this interpretation. If vertical differences reflected only uniform dilution, the three ratios would have been stable. Their significant height-by-period interactions instead indicate differing source distributions, chemical behaviour or transport histories. NH₃ is particularly sensitive to temperature, manure surface conditions and deposition, whereas CH₄ and CO₂ arise from overlapping but non-identical animal and manure processes. The ratios are therefore diagnostic, not substitutes for tracer-based ventilation or mass-balance analysis.

4.3 Wind, neighbouring sources and sensor placement

Westerly periods did not produce a general concentration rise in these data. Their lower internal means suggest that ventilation enhancement dominated during the observed hours,

although this inference remains provisional. A neighbouring naturally ventilated dairy barn could contribute intermittent plumes when upwind, but regional meteorology and internal concentration alone cannot identify that source. A future study should measure approach flow and gas concentrations between the two barns and explicitly test how inter-barn distance, atmospheric stability and plume meandering affect inlet concentrations.

Entropy was response- and height-specific. In Chen et al. (2026), it quantified the sensitivity of simulated airflow variables to prescribed wind cases; here, it quantified temporal occupancy of measured concentration states shaped by sources and ventilation. The values are therefore conceptually related but not directly transferable, and high entropy alone does not identify a universally representative SP.

4.4 Limitations and future work

Sequential sampling trades spatial coverage against simultaneity. Although two-hour complete blocks and mixed models reduced this confounding, rapidly changing flow can evolve within a switching cycle. Campaign and season were also partly aliased with network reduction and analyser deployment. DWD measurements were not co-located at the openings. Finally, retaining every positive concentration avoids an arbitrary physical threshold but makes ratio moments sensitive to small denominators; robust block-level summaries are therefore essential.

Future measurements should deploy three-component velocity sensors at spatial and temporal resolutions matched to the gas measurements, especially at vertical openings and near the roof. The measured concentration fields can then constrain CFD simulations or wind-tunnel experiments that resolve inflow, outflow and recirculation. Such work would permit emission-weighted optimisation rather than concentration-only optimisation, and would test whether the reduced network remains valid across barns, seasons and neighbouring-source configurations.

5 Conclusions

The apparent precision of barn gas concentration measurements depended jointly on gas, spatial coverage and temporal aggregation. Hourly spatial CV was consistently lowest for CO₂ and substantially greater for CH₄ and NH₃; averaging reduced but did not remove this distinction. Concentrations generally increased from bottom to top in autumn and winter, but the ordering was weaker and spatially inconsistent in summer. Accordingly, H1 was supported, whereas H2 was only conditionally supported.

The middle layer at 3.6 m added limited separation during the dense summer observations because it lay only 1 m above the bottom layer. Its omission in Campaign 2 was justified as a design consequence of the observed structure, not as evidence for homo-

geneous mixing. Ratios and entropy maps further demonstrated gas-specific spatial behaviour. For emission applications, concentration sampling should be paired with velocity measurements at matching resolution; until then, reduced networks should be regarded as concentration-representative within the tested barn and periods, not automatically flux-representative.

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Data and code availability

The reproducible R workflow used for the reported analysis is version controlled with the manuscript. Raw instrument data are retained by the authors and can be made available subject to institutional data-management requirements.

A Supplementary meteorological context

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Daily summaries from DWD Potsdam station 03987. Shaded temperature bands show the daily minimum–maximum range. Missing intervals between study periods are intentionally retained.

Figure 8: Daily air temperature, relative humidity and precipitation from the DWD Potsdam station. Monthly ticks run from June to December 2024. Missing intervals between campaigns are retained.

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